The Geography of Market Power: Firm Location, Spatial Rents, and Their Social Costs

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Abstract

The location decisions of spatially competing firms are intertwined, trading off proximity to customers against the cost of intensified competition near rivals, while economies of density across multiple plants add a further complication. We study these economic forces with new data on garage ownership since the 1994 privatization of London's bus market. Exploiting the time dimension together with the exclusive use of garages by single operators, we model locations as choosing firms, rather than the reverse, yielding a tractable estimator despite rich spatial spillovers. Simulations show the highlighted trade-off is economically meaningful: each nearby competitor cuts the private return to capacity expansion by at least 10%. An optimal reassignment of garages yields up to 24.3% in welfare gains, though building restrictions and garage exclusivity leave non-zero local monopoly rents. When also pricing the externalities of empty "dead-mile" travel between garage and route, concentration reduces less, as large operators are best placed to economize on dead miles across their network.

JEL codes: D44, L41, R12, R41

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Where firms locate determines who competes with whom and at what cost consumer demand can be met. This is a central mechanism in industrial organization and urban economics (e.g., Hotelling, 1929; Salop, 1979), yet it remains empirically understudied. Both economies of density across a firm's own plants and competition with nearby rivals generate spillovers across locations that quickly render the firm location problem computationally intractable at any realistic scale.[1] As a result, we know little about how large local monopoly rents firms secure through strategic positioning away from rivals, whether the geographic patterns of private returns to firm location align with social returns, and what role building restrictions on scarce urban land play. We study precisely these questions with new historical data on public bus transportation firms in London and show how features of this market and the data make the location choice problem empirically tractable.

In this market, private bus operators bid for contracts to operate bus routes across Greater London from their own proprietary garages. Operators are less competitive in route auctions for routes far from their garages, due to the *dead mile* costs of transporting empty vehicles between garage and route. On average, supplying 100 minutes of contracted bus service requires 13 minutes of dead mile driving time, with economically meaningful effects on procurement outcomes and firms' bidding behavior. Moreover, when the nearest competing firm is located further away, competition softens so the winning bidder commands higher local monopoly rents in expectation. Besides the costs of dead miles and the benefits of being isolated from competitors, the third spatial profit driver we consider is economies of density from operating routes from multiple nearby garages, which would provide a cost-based explanation of the spatial clustering we document empirically. Henceforth, we refer to these three factors simply as *spatial rents*.

To quantify spatial rents in this market, we exploit the exclusive use of garages by single firms, the scarcity of vacant industrial land in central London—only 2 percent as of 2021, implying a garage stock that is nearly fixed over time—and observing garage ownership changes since the market privatized in 1994. Rather than operators choosing *where to locate*, we interpret the garage ownership data through the lens of a *reversed-choice* framework in which each garage sequentially transfers to its highest-value operator given the state of the network at that point in time. Assuming that garages transactions are competitive and that operators have complete information—motivated by the repeated nature of garage transactions among a small set of firms—the equilibrium transfer price equals the garage's second-highest valuation among all operators, including the incumbent, as in a second-price auction (e.g., Allen, Clark and Houde, 2019; Slattery, 2025). This reversed-choice approach is applicable more broadly when locations are used exclusively by single firms (or plots are small enough that each accommodates only one occupant) and ownership transitions are observed sequentially in the data—conditions that

[1] With F firms and L locations, the number of possible assignments is F^L—with $F = 10$ and $L = 100$, as in our setting, this exceeds 10^{100}, more than the number of atoms in the observable universe. Evaluating even a single firm's entry decision requires integrating over 2^L configurations of rivals' locations.

characterize many markets for commercial real estate, retail space, and industrial land.

Under these conditions, the framework delivers three advantages over standard approaches. First, transaction prices need not be observed: because the equilibrium price is pinned down by the second-highest valuation and enters every operator's indirect utility symmetrically, it cancels from the equilibrium assignment condition—particularly valuable since transaction prices for commercial and industrial property are typically proprietary and difficult to obtain. Second, the framework accommodates unusually rich spatial spillovers across locations without the high-dimensional integration over rivals' strategies that makes standard multi-player entry games computationally intractable, or the moment inequalities that only set-identify structural features of interest. Third, conditioning on the observed network at each transition sidesteps the equilibrium multiplicity that simultaneous location choices across many firms would otherwise generate, yielding a unique prediction for each sequential choice.

The garage value function is grounded in a structural auction model that links spatial positioning to expected profitability through both cost and mark-up channels, while remaining agnostic about competitive conduct at the auction stage. This tight structural link between garage value and expected auction revenues, combined with a small sample of garage transaction data, provides a monetary interpretation of the urban planning counterfactuals.

A parsimonious model specification with only low-dimensional network representations (driving time to route endpoints, closest own garage, number of nearby competitor garages) fits the data remarkably well, achieving a McFadden pseudo-R^2 of 0.554 and a mean AUC of 0.868.[2] The structural estimates reveal that an operator is indifferent between having its closest own garage 3.7 minutes nearer by road and having one fewer competing operator's garage within a ten-minute drive. Beyond spatial positioning, incumbency benefits accumulate steadily with tenure, consistent with the gradual build-up of garage-specific investments and introducing some degree of persistence in the operator-garage network.

To assess robustness to the sequential timing assumption, we estimate a bundle choice specification that treats pairs of consecutive ownership transitions as a joint decision. This introduces a different implicit structural error variance, making direct comparison of coefficients across specifications non informative. Instead, we test equality of coefficient ratios, which are scale-invariant and directly comparable across logit models. A delta method test fails to reject equality of the key trade-off between economies of density and local monopoly rents ($p = 0.945$), while the dead miles coefficient is the one ratio that differs significantly ($p = 0.019$). Taken together, the baseline and bundle estimates establish that the trade-off between proximity to own garages and isolation from competitor garages is the robust central finding of the structural estimation, and that this finding does not depend on the specific timing assumptions of the

[2] *Area Under the Curve (AUC)* is a standard measure in $[0, 1]$ used in evaluating the predictive fit of discrete choice models, summarising how well the estimated utilities rank chosen alternatives above unchosen ones across all operators and time periods.

estimation framework.

We apply the structural estimates to three sets of counterfactuals. We first map the spatial rent surface across all candidate locations in London to identify where the private returns to new garage development are largest. Rather than a monotonic geographic gradient (center to fringe of city), the surface reveals dispersed high-value pockets across the city, reflecting that the 2019 garage network is broadly well-positioned relative to the route network, with the northeast emerging as the main area of genuinely underprovided coverage.

We then use the model to assess the private returns to garage capacity expansion, estimating the value to operators of assigning 50 additional buses to each garage individually. The bus-garage assignment equation underlying these simulations recovers the correct operating garage for 48 percent of routes exactly and 63–72 percent within a two-minute dead-mileage tolerance against observed tender data. Regressing estimated expansion values on structural covariates reveals that the dominant cost driver is the average dead miles of currently assigned routes, while incumbency tenure and spatial isolation from competitors — the local monopoly rent channel — are the main drivers of heterogeneity across garages. One additional competitor garage within a 10-minute drive reduces the return to capacity expansion by at least £0.9M, approximately 10 percent of the baseline expansion value. This quantifies a classic divergence between private and social returns to infrastructure investment (Laffont and Tirole, 1993): the socially desirable margin of capacity investment would minimize dead miles, reducing procurement costs borne by the taxpayer and the environmental burden of empty bus travel in a congested city, whereas market power considerations generate conflicting incentives.

Finally, we use the structural estimates to evaluate the welfare gains from reassigning existing garages across operators under a social planner's objective. Reducing spatial isolation between competitors is the dominant source of gains, contributing up to 24.3% in the unconstrained long-run scenario. Pricing dead-mile externalities—pollution, noise, and congestion—barely changes welfare gains, but concentration falls by less under this correction as large operators are best placed to economize on dead miles across their own network, and the planner tilts reassignment toward them rather than toward only dispersing rivals. Under a short-term bus fleet constraint, garage holdings remain more concentrated as small firms expand less, reducing the welfare gain from eliminated local monopoly rents to 19.6%. When we further account for the social costs of writing off incumbent-specific garage investments—captured by a conservative linear specification based on the structural estimates—welfare-improving transactions remain feasible despite this specification strongly favoring the status quo.

Our empirical approach builds on a literature that uses endogenous firm location decisions to learn about spatial rents, market structure, and competition (e.g., Mazzeo, 2002; Seim, 2006; Jia, 2008; Holmes, 2011; Ellickson, Houghton and Timmins, 2013; Nishida, 2015; Zheng, 2016; Oberfield et al., 2024). The closest antecedents are Jia (2008) and Nishida (2015), who estimate location choice games with both economies of density and competitive effects, and Holmes (2011)

and Houde, Newberry and Seim (2023), who quantify economies of density in distribution networks using a moment inequality approach. Our setting — many firms, exclusive-use locations, a congested mega-city, and regulated fixed demand — permits identification of an unusually rich set of spatial rents: economies of density, local monopoly rents from strategic competitor positioning, and transportation costs simultaneously with a simple estimation strategy.[3] Relatedly, Cao et al. (2026) provide a framework to address potential distance endogeneity in location choice models; this concern does not arise here as both garage plots and "demand" (expected lump-sum payments from Transport for London) are fixed.

Our setting connects naturally to an urban economics literature on agglomeration and the spatial organization of cities (e.g., Combes and Gobillon, 2015; De Palma et al., 2011). Heblich, Redding and Sturm (2020) show that railway infrastructure shaped workplace-residence separation and land values in London through agglomeration forces; we study the complementary question of how industrial land frictions and private firm location choices generate market power within the city's existing bus route network. The urban policy counterfactuals connect to a growing literature on transport network design (Fajgelbaum and Schaal, 2020; Allen and Arkolakis, 2022; Balboni, 2025; Kreindler et al., 2023) and heterogeneous returns to infrastructure investment (Brancaccio, Kalouptsidi and Papageorgiou, 2024). We identify where private and social returns to garage investment diverge, with a single source of a wedge: spatial isolation from competitors generates local monopoly rents that push operators away from the cost-minimizing configuration. Through this mechanism, private garage ownership transfers rents from the public (taxpayers) to firms, increasing dead miles and consequently inflating congestion, noise, and exhaust emissions beyond what cost minimization alone would require. This connects to the classic urban economics of transport investment (Small, Verhoef and Lindsey, 2024) and to the Pigouvian tradition (Pigou, 1920; Laffont and Tirole, 1993) of identifying market failures from uninternalized externalities.

The garage value function is grounded in a structural auction model (Guerre, Perrigne and Vuong, 2000; Cantillon and Pesendorfer, 2006, 2007). Our analysis of location choice externalities thus offers a spatial interpretation of a well-established finding in procurement design: when quality has noncontractible dimensions, awarding contracts on price alone generates adverse selection and distorted investment incentives (Laffont and Tirole, 1986; Bajari and Tadelis, 2001; Lopomo, Persico and Villa, 2023; Decarolis, 2014; Gagnepain and Ivaldi, 2002).[4]

[3] For example, Holmes (2011) abstracts from local monopoly rents and economies of density. On the other hand, the moment inequality approach delivers bounds on the part of fixed costs that varies across locations, which is not identified in the reversed-choice framework. We discuss this limitation when simulating counterfactual (urban planning) policy scenario's.

[4] *Transport for London* procures bus routes on price alone conditional on contractible service attributes, leaving garage location—the primary margin on which operators can extract rents above cost minimization—outside the contractual framework. In Marra and Oswald (2024), we highlight this and document that the operational efficiency of the bus-garage assignment can improve by 14%, looking at dead miles alone. The current paper focuses on the strategic forces governing the garage ownership network itself.

In Appendix C, we assess a complementary interpretation of the spatial clustering of operators: whether it partly reflects collusive market-sharing, a concern the UK Competition Commission (2011) raised for local bus markets. We exploit the structural link between garage location values and expected procurement profits to construct implied punishment values for profitable transactions that did not occur. This relates to the literature on collusion detection in procurement (e.g., Porter, 1983; Porter and Zona, 1993, 1999; Bajari and Ye, 2003; Kawai and Nakabayashi, 2022)—which typically takes firms' assets and locations as given when testing for bid-level collusion—and to work on how market structure and firm positioning can facilitate tacit coordination (e.g., Miller and Weinberg, 2017; Bourreau, Sun and Verboven, 2021; Berry and Waldfogel, 2001; Sullivan, 2017). Unrealized profitable transactions concentrate on spatially isolated garages, reinforcing the private–public wedge in location incentives; their incidence does not, however, decline with the number of active firms as collusion theory predicts, and we take no stance on whether market-sharing dynamics underlie part of the observed garage network.

The remainder of the paper is organized as follows. Section 1 describes the market and data. Section 2 presents the structural model. Section 3 presents the estimation results and robustness checks. Section 4 presents the urban policy counterfactuals. Section 5 concludes.

1 Industry and data description

London Bus Services Ltd (London Buses) is part of Surface Transport within *Transport for London (TfL)*, in charge of delivering the Mayor's Transport Strategy. This involves transporting over six million passengers on 7,700 scheduled buses across 675 different routes stretching over 490 million kilometers of road per year. Procuring bus operation services from private operators costs TfL on average £273m annually over the period 2003–2019, paid for by UK taxpayers. The market has been organized in this way since it was privatized along geographic lines in 1994. Today, six large international transportation firms and a handful of smaller operators compete for tenders.[5] This section describes the market structure and our data, documents the spatial clustering of bus garages, and provides reduced-form evidence linking garage network position to route auction outcomes.

For our analysis of spatial rents, we constructed a comprehensive dataset of the operator and bus garage network in London between 1994–2019. Each operator maintains its own proprietary garage network, and the location of these garages *vis-à-vis* the routes they service is crucial: operators are less competitive in route auctions for routes far from their garages, due to the dead mile costs of transporting empty vehicles between the garage and the route endpoints. As a result, the distance of an operator's garage to the bus route in question has an economically meaningful effect on procurement outcomes and firms' bidding behavior, as we document in

[5]The major operators over our sample period are Abellio, Arriva, Go-Ahead, Metroline, RATP, and Stagecoach.

Section 1.2 below. Throughout, we refer to the firm operating buses from a garage as the garage *owner*—regardless of whether the firm holds a freehold or long leasehold contract—and we exclude the small number of cases where a garage is shared between two operators. Figure A3 documents changes in garage ownership over time by operator in our sample.

The London Omnibus Traction Society provided a historic dataset based on their archive of bus schedules, matching the majority of routes tendered since 2005 to the garage from which they were operated, and containing information about garage openings and closings. We complement this with garage location information from the privately-run London Bus Routes website, bus stop locations from the TfL Open Data Initiative, and garage addresses verified using the Google Maps API. We use the actual driving time as our preferred measure of distance to fully account for the spatial nature of the problem, including geographic obstacles such as the River Thames that can only be crossed at certain points. For a comprehensive description of the market and the data, see Marra and Oswald (2024).

A final dataset contains 2017 cross-sectional median house price aggregates obtained from Land Registry Price Paid Data, aggregated to the 33 Greater London local authorities (the 32 boroughs plus the City of London) and the 679 Greater London electoral wards HM Land Registry (2018). Matching each garage to its borough and its ward, this provides local land-value controls at two spatial resolutions—the finer ward series, and a coarser borough series that is more robust where ward-level transactions are sparse—used to address potential confounds in the estimated return to capacity expansion in Section 4.2.

The structural estimates are based on *movers*, e.g. changes to the ownership of garages observed over time. The estimation sample is based on garage-operator network changes taking place after 1994. We hus exclude the first year when the market was being privatized and when the garage network was being formed, as the initial privatization assignments are not strategic choices in the model's sense. We start with 112 garages and exclude three garages with incomplete information about the ownership history between 1995-2019, seven garages that were shared at some point in time, and one temporary garage. Finally, 10 of the remaining 101 garages never changed ownership in our data, so the estimation is based on (multiple) ownership changes in the remaining 91 garages.

These 91 garages account for a total of 192 ownership changes, which form the basis of our structural estimation. Table 1 lists the type of ownership change that took place. In 42 percent of the times that the garage-operator network changes, a garage is acquired from another operator. In 32 percent of the time the network changes because an operator occupies a garage for the first time. In the remaining cases the network changes because a garage is closed down (33 times) or temporarily not used to operate bus routes (17 times). We interpret these latter cases as the garage being bought from or sold to *the market*, as described further in the theoretical framework. Furthermore, the relevant unit of time will be a so-called *changedate*, which we define as a date at which we observe any change in the garage-operator ownership structure. There

Table 1: Summary of all garage network changes 1995-2019

Type of change	Count	Share
Closed down	33	0.172
New entry	61	0.318
Ownership change	81	0.422
Temporarily vacant	17	0.088
Total	192	1

Number	Share by change-date	
	Operators changing garage	Garages entered by operator
1	0.657	0.756
2	0.143	0.098
3	0.043	0.024
4	0.029	0.037
5	0.000	0.012
6	0.100	0.049
8	0.014	0.012
9	0.014	0.012

Notes: The left panel accounts all the changes in the garage-operator network observed in the data. The right panel documents the number of operators entering a garage and the number of garages entered by an operator simultaneously, that is, on the same date. The right panel is based on all times when operators opened a new garage (the 61 "New entry" events) or purchased an existing garage from another operator (the 81 "Ownership change" events) between 1995-2019.

are 114 changedates in our data from 1995 to 2019, with on average 1.68 ownership changes per changedate. In 66% of cases there is a single operator changing garage on a given changedate. The vast majority of network changes (76%) also involve just one garage per time. These statistics are documented in Table 1 and exploited in the estimation strategy.

To best capture spatial rents associated with the geographic location of garages relative to routes and other garages, distances are measured as the exact drive times on the London road network.[6] We use the average drive time to the two endpoints of a route as our primary dead miles measure (*Dead Miles (Start-Stop Minutes)*). This measure reflects that a route needs to be serviced from both directions, so being near only one endpoint is not necessarily optimal. A garage at the midpoint of a long route, for instance, minimizes total dead miles even if it is not adjacent to either terminus. We consider the sensitivity of our results to other dead mile measures, including the distance to the nearest route endpoint and the average distance to a stopover point where at least 5 routes intersect in Sections 1.2 and 3.4.

In the remainder of this section, we document new facts about the observed garage-operator network and its relation to bidding in route auctions.

[6]Simple straight-line distance between points would miss a great deal of nonlinearities arising from geography and, ultimately, true transportation costs. We therefore use the Open Source Routing Machine (OSRM) to compute optimal drive times between all garages and all other garages and all 55,578 bus stops in London. Besides the distance to the endpoints of the routes, we also consider the distance to the closest stopover point, where at least 5 routes intersect. See Marra and Oswald (2024) for further details.

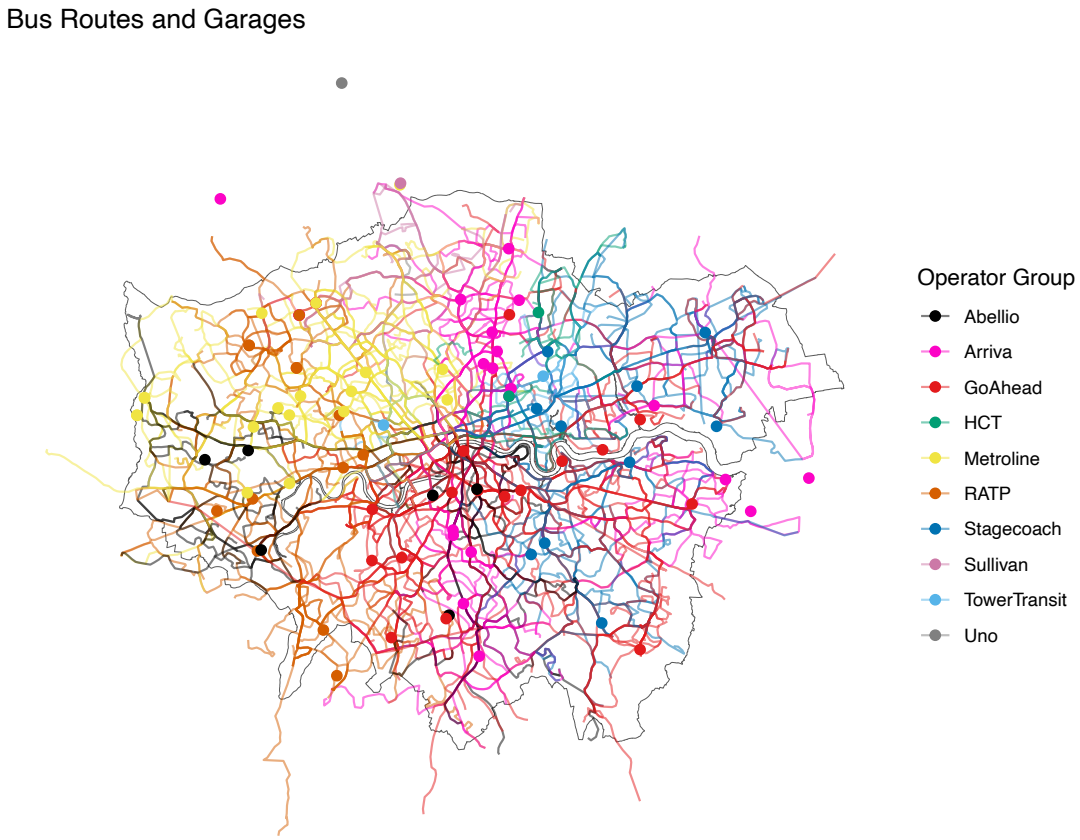


Figure 1: Route network and garage locations by operator group.

Notes: Garage-route network at the end of the sample (2019). Points denote locations of garages, line segments indicate bus routes, and colors indicate which operator owns the garage or operates the route.

1.1 Spatial clustering of garages

Figure 1 shows a snapshot of our dataset in 2019, which links routes to the garage where they were operated from. A key observation is the spatial clustering of bus garages. Specifically, the figure shows that garages of a certain operator group are concentrated in certain areas of London, rather than being uniformly spread out across the city. The structural model and counterfactual simulations are designed to understand the factors driving this observed spatial clustering. The model is based on changes in the garage ownership network over time. Garages change ownership between 1–7 times, and no obvious spatial pattern can be detected as to which garages change ownership more frequently (Appendix Figure A1). This makes systematic selection of peripheral versus central garages into the estimation sample unlikely.

One explanation for operating routes from nearby garages is the specialized functions that some garages perform within an operator's network. Driver training, maintenance, engineering, logistics, and HR are among the functions that are often centralized in the larger garages.[7]

[7]Table A1 in the Appendix lists the specialized functions performed by select garages within operators' net-

Table 2: Route auctions and Garage Networks of winning bidders

	N	Mean	SD	Min	Max
Accepted Bid (Million Pounds)	1907	2.80	2.28	0.00	18.17
Number of Bidders	1907	2.79	1.10	1.00	9.00
Route Length	1846	7.88	2.56	2.00	24.00
Peak Vehicle Requirement (PVR)	1850	12.45	7.76	0.00	53.00
Dead Miles (Start-Stop Minutes)	1457	13.28	5.10	2.55	40.45
Dead Miles (Start-Stop km)	1457	7.78	3.45	1.38	26.62
Dead Miles (Stopover Minutes)	1333	11.24	6.18	0.83	48.80
Dead Miles (Start Minutes)	1476	12.27	7.58	0.00	46.00
Closest Own Garage (mins)	1492	12.20	10.65	0.00	48.80
Dead Miles of Closest Competitor (mins)	1779	13.33	4.91	3.10	33.45
Total Number of:					
Auctions	1907.00				
Auctions w/ observed dead miles	1457.00				
Unique Routes	698.00				
Routes w/ observed operating garage	592.00				

Notes: Summary statistics for variables used in the route auction analysis in Table 3. *Dead Miles* denote driving time from the winner's garage to a spatial point on the auctioned route. *Start-Stop* is the average distance to route start and end points; *Stopover* refers to a bus stop with at least 5 route intersections (95th percentile). Sample sizes vary due to incomplete garage assignment (592 of 711 routes observed) and route characteristics (Route Length, PVR) from Wikia and LBR data sources.

Together with the capacity constraints due to building restrictions in the densely populated city, these might generate cost synergies from operating routes from nearby garages. For instance, a report of the UK Industrial Land Commission documents that in 2021 (close to the end of our sample period), only 2 percent of the industrial land in central London is left vacant, constraining expansion opportunities of existing garages.[8]

1.2 Link between garage network and revenue from route operation

We next relate winning bids from the publicly available TfL tender data to our network variables to support the claim that the placement of a garage in the network of routes and other garages is relevant for route auction outcomes. We observe the operating garage for 592 out of 698 routes; the remainder are cases where the operating garage was not a mandatory field in the tender documentation and was left unreported. To the extent that garages are more likely to be reported when they are closer to the route—and thus more strategically salient—this selection could bias the reduced-form estimates. The structural estimates of Section 2 are not subject to this concern, as they are based on garage proximity and capacity measures that are universally observable and formalized in the assignment equation (5). We subsequently compute the empty driving

works, based on information on their websites.

[8]See https://centreforlondon.org/wp-content/uploads/2022/01/CFL-IndustrialLand-v4-1.pdf.

Table 3: Explaining the size of accepted bids for London bus routes

	Accepted Bid (in Million Pounds Sterling)				
	Start-Stop			Start	Stopover
	(1)	(2)	(3)	(4)	(5)
Dead Miles	0.099***	0.019**	0.005	0.003	0.009+
	(0.011)	(0.006)	(0.007)	(0.004)	(0.005)
Route Length	0.067**	−0.013	−0.022+	−0.022+	−0.024+
	(0.021)	(0.012)	(0.012)	(0.012)	(0.013)
Number of Bidders		−0.149***	−0.126***	−0.125***	−0.126***
		(0.029)	(0.029)	(0.029)	(0.032)
Peak Vehicle Requirement (PVR)		0.237***	0.225***	0.226***	0.225***
		(0.004)	(0.005)	(0.005)	(0.005)
Closest Own Garage (mins)			−0.006+	−0.006	−0.007+
			(0.004)	(0.004)	(0.004)
Dead Miles of Closest Competitor (mins)			0.043***	0.044***	0.045***
			(0.008)	(0.007)	(0.008)
Constant	0.172	−0.786***	−0.875***	−0.874***	−0.913***
	(0.332)	(0.207)	(0.223)	(0.223)	(0.246)
Year FE	✓	✓	✓	✓	✓
Winner FE	✓	✓	✓	✓	✓
Num.Obs.	1,457	1,457	1,457	1,457	1,307
R2	0.196	0.748	0.754	0.754	0.748

Notes: Statistical significance: + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001. Dead Miles in all models equals driving time in minutes between the winner's garage and certain spatial characteristic of the auctioned route: Columns (1)-(3) use average distances to start and end points of the route, column (4) uses distance to route start point only, column (5) uses distance to stopover points. See Table 2 for variable details.

time for 1,457 out of 1,907 route-auctions. We regress the winning bid for a route contract on: i) auction and route characteristics, ii) our computed measures of the dead miles driving time between the route and the garage where the route is operated from, iii) the driving time between the operating garage and the closest garage of the same operator (*Closest Own Garage*, to capture economies of density), and iv) the dead miles of the closest competitor garage (to capture local monopoly rents).

On average, buses drive 13.28 minutes (or 7.78 km) empty each time they move between garage and route, as measured by the *Dead Miles Start-Stop* variable (see Table 2 for summary statistics). An additional minute of dead mile driving time increases the winning bid by at least £19,000, as shown in columns (1)–(2) of Table 3; the estimate is larger when excluding auction-level controls for the Peak Vehicle Requirement and the number of bidders, which absorb part of the variation in bids. Adding the strategic network variables in column (3) renders the dead miles coefficient insignificant. This pattern is consistent with dead miles affecting both costs—pushing bids up—and competitive positioning—potentially compressing mark-ups as operators near a route also tend to face more local competition. These offsetting forces are difficult to separate in a reduced-form regression on winning bids. The structural model of Section 2 identifies

the net effect on operator profits rather than on winning bids, which sidesteps this issue.

The results also point to a negative correlation between the accepted bid and the proximity of the winning bidder's *Closest Own Garage* to the operating garage, consistent with cost synergies from a spatially concentrated garage network. The coefficient on this variable should be understood to reflect a mix of cost effects and mark-up adjustments from being better positioned than competitors. The effect of competitor proximity, by contrast, is more cleanly interpretable: since the drive time to a competitor's garage does not directly affect the winning operator's costs, this coefficient plausibly reflects the competitive effect on equilibrium mark-ups alone. The winning bid is about £43,000 higher for each additional minute of drive time between the operating garage and the nearest competitor garage, consistent with operators earning higher mark-ups when shielded from nearby rivals.

The results are robust to alternative dead mile measures, as shown in the final two columns of Table 3. The distance to the closest route endpoint (column labeled "Start") is insignificant in the full specification. The distance to the nearest stopover point —where at least five routes intersect— (column labeled "Stopover") enters with the expected sign but is statistically weaker and economically smaller, with each additional minute increasing the winning bid by £9,000. The other coefficients are also stable across distance specifications.

2 Model

This section develops a static two-stage model that links the value of a firm's location to the spatial rents it can obtain, in order to estimate the determinants of those rents from observed location choices. Time is divided into periods. At the beginning of each period, operators observe the current garage-operator network and draw idiosyncratic cost shocks for each route. In stage 1, changes to the garage-operator network occur. In stage 2, operators bid on bus route contracts given the updated network.[9] Firms are myopic with respect to future changes in the garage-operator network, but persistence induced by five year route contracts is accounted for in how much they value garages.[10] The process then repeats in the next period.

[9]Contracts typically start 8 to 10 months after the award date so that, where necessary, winning bidders can reorganise their logistics and obtain additional buses. In figure A7 we verify empirically that in the vast majority of cases, operators obtain first the garage, and then bid for new routes.

[10]Location persistence operates through the garage network, and not the route network, since reliable (tender) data on who operates which route is only available from 2004 onwards. We assess the implications of omitting further dynamics in Section 3.2 when considering one-period ahead decisions.

2.1 Stage 2: Bidding for routes given a garage network

In the second stage, firms compete for route contracts in a first-price sealed bid auction given the existing operator-garage network Γ, which is public information.[11] The cost for operator i to operate route r from garage j can be written as

$$c_{ijr}(\Gamma) = \tau\big(\delta_{ijr}(\Gamma), \nu_{ijr}\big) \tag{1}$$

where $\delta_{ijr}(\Gamma)$ collects the deterministic cost components: route-specific variables such as service length and frequency, and factors related to the position of garage j in the network of routes and garages Γ, including dead miles between the garage and the route and, if economies of density are present, the distance between garage j and other garages owned by operator i.[12] The term ν_{ijr} captures idiosyncratic unobserved cost variation that is private information to the firm, varying independently across operators, routes, and garages, with $\tau(\cdot)$ increasing in $\nu_{ijr} \sim F_\nu$.

Operators participating in the auction bid to maximize expected route profits,

$$\pi_{ijr}^{route}(\Gamma) = (b_{ijr} - c_{ijr}(\Gamma))\, P(\text{win}|b_{ijr};\Gamma) \tag{2}$$

where $P(\text{win}|b_{ijr};\Gamma)$ is the probability of winning the auction when submitting bid b_{ijr}, given the operator-garage network Γ. To accommodate both competitive and non-competitive explanations for the observed spatial clustering of garages, we do not impose that all potential bidders submit competitive bids. We only impose the minimal restriction that lower-cost firms achieve higher expected procurement profits in equilibrium. This assumption holds for the unique Bayes-Nash equilibrium in the competitive case, where the setting reduces to a standard independent private values first-price procurement auction.[13] It also holds for a strong cartel with side-payments and, by the results of Pesendorfer (2000), for a weak cartel without side-payments when sufficiently many contracts are auctioned, as both cases yield (approximately) efficient allocations. This assumption therefore accommodates a wide range of strategic be-

[11] While the first-price auction mechanism simplifies the exposition, TfL awards route contracts via combinatorial auctions where firms may submit route package bids. Abstracting from package bids is without loss of generality. Moreover, Cantillon and Pesendorfer (2007) find limited empirical evidence for cost synergies in operating multiple routes at once, given garage ownership, possibly because garage capacity constraints may limit the cost savings from operating more routes at once. Our empirical results point to economies of density at the garage level, consistent with this interpretation.

[12] In practice, operator i operates route r from its closest garage; we are more precise about this in Section 2.2.

[13] The unique symmetric Bayes-Nash equilibrium $\sigma^*(c_{ijr})$ solves (Guerre, Perrigne and Vuong, 2000):

$$\sigma^*(c_{ijr}(\Gamma)) = c_{ijr}(\Gamma) + \frac{P(\text{win}|\sigma^*(c_{ijr}(\Gamma));\Gamma)}{P'(\text{win}|\sigma^*(c_{ijr}(\Gamma));\Gamma)} \tag{3}$$

under standard regularity conditions, where $P'(\text{win}|\sigma^*(c_{ijr});\Gamma)$ denotes the derivative of $P(\text{win}|b_{ijr};\Gamma)$ with respect to the bid. The mark-up decreases in the number of competitors and is compressed when cost distributions are less dispersed, for instance when all operators have garages at similar distances from the route.

havior at the auction stage, including bidding cartels with unspecified internal organization.[14] Where multiple equilibria exist, we assume that players coordinate on a symmetric bidding strategy $\sigma^*(\cdot)$, and denote the resulting expected equilibrium profits by $\pi^{*route}(\cdot)$.

Economies of density and dead miles affect $\pi^{*route}(\cdot)$ directly through costs, but also generate local monopoly rents through the equilibrium mark-up. To illustrate this, consider the win probability in the IPV first-price auction with additively separable $\tau(\cdot)$:

$$P(\text{win}|b_{ijr};\Gamma) = \prod_{h\neq i}\left[1 - F_v\left(\sigma^{-1*}(b_{ijr}) - \delta_{hj'r}(\Gamma)\right)\right] \tag{4}$$

where j' is the garage from which operator h would operate route r. Conditional on b_{ijr}, the probability of winning increases in the distance between competitors' garages and the route, since greater distance implies higher deterministic costs $\delta_{hj'r}(\Gamma)$ for competitor h. Costly dead miles therefore generate local monopoly rents, particularly for operators whose garages are more isolated from those of their rivals. Symmetrically, operating from a denser own garage network might lower costs and raise the probability of winning, further increasing expected profits. Having established how garage location drives expected rents from TfL route auctions, we next turn to what this implies for the garage-operator match in the first stage, where $\pi^{*route}(\cdot)$ serves as the value function that governs location choices.

2.2 Stage 1: Garage Entry

The value of garage j to operator i, Π_{ij}, is the sum of expected profits generated through route procurement across all routes the garage can service. Given the role of capacity constraints in the London bus market, we must first determine the set routes which are served from each garage. Let Capacity_j denote the number of buses that garage j can accommodate, and let PVR_r denote the Peak Vehicle Requirement of route r—the required number of buses needed to operate the route at peak demand, as mandated by TfL. Buses are allocated to the garage starting from the nearest routes, until the capacity constraint binds. Specifically, letting $r_j^{n:R}$ denote the nth closest out of R routes to garage j, the sequence of routes ordered by increasing dead miles to garage j is

$$\mathbf{r}_j = \{r_j^{1:R}, r_j^{2:R}, \ldots, r_j^{R:R}\} \tag{5}$$

with $r_j(m)$ denoting the mth route in this sequence and $PVR_j(m)$ the corresponding PVR. We maintain that the profit function exhibits constant returns to scale in PVR, so that each bus

[14] For example, some firms may form a bidding ring and hold an internal knockout auction to determine their representative bidder in the target auction. In that case, (2) applies to the representative ring member and the competitive bidders, and side-payments must be made conditional on winning to other ring members (see e.g., McAfee and McMillan, 1992; Pesendorfer, 2000; Asker, 2010).

stationed at a garage contributes equally to the value of that garage.[15]

Garages are more valuable to operator i when they have greater capacity, are located closer to more profitable routes, closer to other garages owned by i, and further away from competitors' garages. The garage value function that captures these forces is

$$\Pi_{ij}(\Gamma) = \sum_{m=1}^{R} \sum_{v=1}^{PVR_j(m)} \beta_{r_j(m)} \frac{\pi^{*route}_{ijr_j(m)}(\Gamma)}{PVR_j(m)} \mathbf{1}\left(\sum_{l=0}^{m-1} PVR_j(l) + v \leq \text{Capacity}_j \right) \tag{6}$$

with $PVR_j(0) = 0$, $\mathbf{1}(\cdot)$ the indicator function, and $\beta_{r_j(m)}$ the discount rate associated with the timing of the auction for the mth closest route. The fraction times β_r gives the present value of expected profit per bus stationed in garage j from the procurement contract for route $r = r_j(m)$. The summation over routes and buses (vehicles v), truncated by the indicator function, accounts for filling the garage to capacity with buses from routes in order of increasing dead miles.[16] Distance from competitors' garages generates local monopoly rents through the equilibrium bid mark-up, while proximity to routes and own garages affects variable profits through operating costs.

Two considerations motivate a semi-structural rather than fully structural approach to estimating (6). First, while the route auctions are modeled independently for tractability, TfL in practice auctions routes in small batches using package bidding. Identifying the joint cost distribution in a combinatorial auction model requires observing all bids for each route (Cantillon and Pesendorfer, 2007), which is not feasible as the available data contains only the winning and lowest bids. Second, taking the garage value function directly to data requires taking a stand on competitive conduct at the auction stage. Assuming competitive bidding would preclude the assessment of whether collusive market-sharing explains the observed spatial clustering, while characterizing the auction equilibrium under collusion requires strong assumptions about the cartel's internal organization that are not warranted without further evidence.

We therefore follow the standard convention of the entry literature and adopt a semi-structural approach (e.g., Seim, 2006; Jia, 2008; Holmes, 2011; Ellickson, Houghton and Timmins, 2013), characterizing Π_{ij} as a function known up to a vector of parameters but informed by the structural representation in (6). Modeling the auction stage explicitly serves two purposes even

[15]This assumption is supported by the operational structure of the London bus market, where individual buses operate on fixed schedules independently monitored by the firm's operations center, with no material scale economies at the vehicle level. For example, if a bus is running late on its own schedule, the operations center controllers contact the driver to take action, as explained on this forum: *"Every route has time cards, you leave at a specific time and have multiple way points where you are supposed to be at a predetermined time. London zoo at 9.27, Baker street at 10.22 for example. If you run too late behind schedule then the controllers will instruct you to change the destination board, terminate early and give passengers transfer tickets. You turn the bus at a specific point and then hopefully you're on time heading the opposite direction again."*.

[16]The garage value function naturally implies that operators only exert competitive pressure on routes not too far from their garages when the effect of dead miles on operating cost is sufficiently large, so that the model does not require an ad hoc distance cutoff in bidding.

within this semi-structural framework. First, it disciplines the choice of network variables by separating those that operate through costs—such as dead miles and own garage proximity—from those that operate through equilibrium mark-ups—such as competitor garage proximity—as illustrated in Section 1.2. Second, it supports the counterfactual welfare calculations in Section 4, where we link estimated location utilities to changes in procurement costs.

We now describe how the network Γ is formed. A salient feature of the London bus garage market is that garage use is exclusive: the incumbent operator must be willing to vacate a garage before another firm can enter. In addition, building constraints in London mean that the set of garages is nearly constant over time. These two features—exclusive use and a stable garage stock—allow us to recast the entry game as one where *each garage choose one operator*, rather than operators choosing a *set* of garages. As we show in the next section, this reversal delivers a tractable multinomial estimation framework and avoids the need to solve explicitly for operators' location strategies. Moreover, since all potential entrants must pay the market garage price, prices cancel in the garages choose operators framework, and we do not need to observe them to estimate the model, as formalized below.

2.2.1 Garages choose operators

We assume complete information at this stage. This is motivated by the repeated nature of interactions among the same set of firms across all garage transactions: operators are well-informed about each other's valuations, and buyout decisions involve extended negotiation rather than simultaneous moves. Under complete information, the garage price determination process can be described as a second-price auction, as in other negotiated-price asset markets (Allen, Clark and Houde, 2019; Slattery, 2025): the winning operator pays a price equal to the value of the garage to its second-most-interested buyer, since the incumbent will not sell below that threshold and competing operators will not bid above their own valuation.

The equilibrium price of garage j is therefore

$$p_j^* = \Pi_j^{N-1:N} \tag{7}$$

where $\Pi_j^{N-1:N}$ denotes the second-highest value of garage j among all N market participants, with $\Pi_{ij}(\Gamma)$ reflecting the expected profits to operator i in the hypothetical scenario where i enters garage j, holding the rest of the operator-garage network fixed. All potential entrants face this same price: $p_{ij} = p_j^*$. Importantly, the incumbent operator is included in this order statistic, since the highest-value entrant can only acquire the garage at a price exceeding the incumbent's own valuation. The transaction price thus serves as an endogenous opportunity cost of retention for the incumbent, analogous to a second-price auction with the seller as an additional bidder. Since transaction prices enter the indirect utility functions of all operators symmetrically, they cancel in equilibrium, and we do not need to observe them to estimate the model. We further

assume that operators are driven by their full variable profits in this price negotiation process (including the idiosyncratic component ϵ_{ij}), supporting that at least the identity of the highest-value operator for each garage is common knowledge. This is consistent with the repeated and transparent nature of garage transactions among a small set of firms.

To formally characterize the equilibrium of the garage-chooses-operator game, the payoff for garage j under action $a_j = i$ is the value to operator i net of the transaction price:

$$U_j(a_j, \mathbf{a}_{-j}) = \Pi_{ij}(\Gamma) - p_j^*(\mathbf{a}_{-j}) \tag{8}$$

where we write $p_j^*(\mathbf{a}_{-j})$ to make explicit that the equilibrium transaction price depends on the actions of all other garages through their effect on the network Γ. Since $p_j^*(\mathbf{a}_{-j})$ is determined by the second-highest valuation among all operators, the winning operator — whose dominant strategy in the second-price auction is to bid their true valuation — takes this price as given and cannot influence it by their own action. The transaction price therefore cancels from the best-response correspondence, and the equilibrium assignment reduces to

$$a_j^* = \arg\max_{i \in \mathcal{N}} \Pi_{ij}(\Gamma) \tag{9}$$

so that each garage is assigned to the operator with the highest value for it, and transaction prices need not be observed to estimate the model.[17]

The set of garages choosing an operator is C (with cardinality C). Each garage chooses an operator from the set $\mathcal{N}$ (with cardinality N) of active operators, with $a_j = i \in \mathcal{N}$ indicating the selected operator. The vector $\mathbf{a} \equiv \{a_j : j = 1, \ldots, C\}$ collects the actions of all garages, and $\mathbf{a}_{-j}$ those of all garages other than j. The Nash equilibrium of this game is the C-tuple $\mathbf{a}^*$ such that for every garage j:

$$a_j^* = \arg\max_{i \in \mathcal{N}} U_j(i, \mathbf{a}_{-j}^*) \tag{11}$$

A pure-strategy Nash equilibrium exists when the garage value function $\Pi(\cdot)$ satisfies supermodularity—so that operator i entering garage j increases i's value for all other garages that become closer to its network—and decreasing differences—so that the entry of an additional competitor within proximity of garage j reduces its value to operator i. Under these conditions,

[17] The cancellation follows from $p_j^*(\mathbf{a}_{-j})$ not varying with i in the maximisation:

$$a_j^* = \arg\max_{i \in \mathcal{N}} U_j(i, \mathbf{a}_{-j}) = \arg\max_{i \in \mathcal{N}} \left[\Pi_{ij}(\Gamma) - p_j^*(\mathbf{a}_{-j})\right] = \arg\max_{i \in \mathcal{N}} \Pi_{ij}(\Gamma) \tag{10}$$

Note that economies of density make garages potential complements for the same operator, which by Milgrom (2000, Theorem 4) could preclude the existence of a competitive equilibrium price *vector* across all garages simultaneously. This concern does not arise here because prices are determined bilaterally for each garage via (7): existence of p_j^* requires only a well-defined second-highest valuation for each garage, which holds under the maintained assumptions on Π_{ij}. The same argument applies to bundle transaction prices in Section 2.3.2.

Table 4: Summary statistics, location choice model

	Mean	SD	Min	Max
No. bus route starts within 10 min. drive time (RouteStarts10Min)	14.15	7.09	0.00	32.00
Min. drive time to any own garage (MindurOwnGar)	27.94	17.02	0.00	86.40
No. own garages (NrGarages)	6.24	5.23	0.00	19.00
No. comp. garages within 10 min. drive time (CompGar10Min)	1.75	1.59	0.00	7.00
No. bus route stopover points within 10 min. drive time	33.55	25.22	0.00	139.00
Avg. drive time to any own garage	38.67	14.54	0.00	86.40
Avg. drive time to any comp. garage	38.86	5.99	28.32	57.89
Years owned by previous operator (YearsInc)	7.19	5.96	0.00	25.53
Alternative is selected	0.09	0.28	0.00	1.00
Garage is not vacant	0.90	0.31	0.00	1.00

Notes: Summary statistics for variables used in the garage-operator utility model in equation (12) (top panel), for additional variables including alternative distance measures (middle panel), and for other characterizations of the discrete choice model (bottom panel). The top two panels are based on all 1,649 observations (ijt) in the estimation sample where the garage is non-vacant; summary statistics for the bottom two variables are based on the full estimation sample of 1,841 observations.

a pure-strategy Nash equilibrium exists and can be computed via an iterative best-response algorithm (Topkis, 1978, 1979). Multiple equilibria generally exist in entry games of this type (see Aguirregabiria and Suzuki, 2016). Following, e.g., Jia (2008) and Nishida (2015), we compare estimates across equilibria to assess robustness; details are provided in the next section.

2.3 Estimation and identification

We specify a functional form of the garage utility function that is additive in observables and a residual error term:[18]

$$\begin{aligned}\tilde{\Pi}_{ij}(\Gamma) = & \underbrace{\theta_1 \text{YearsInc}_j + \theta_2 \text{Nonvacant}_i + \theta_3 \log(\text{NrGarages})_i}_{X_{ij}'\theta} \\ & + \underbrace{\theta_4 \text{RouteStarts10Min}_j + \theta_5 \text{MindurOwnGar}_{ij} + \theta_6 \text{CompGar10Min}_{ij}}_{G_{ij}'\theta} + \epsilon_{ij}; \end{aligned} \tag{12}$$

where θ collects the parameters of interest—principally those governing economies of density, local monopoly rents, and incumbency benefits—X_{ij} contains observed characteristics of the garage and operator, and G_{ij} contains network variables describing: the centrality of garage j in the exogenous route network; the centrality of garage j in competitors' garage networks

[18] The structural estimates are in units of utility; their monetary interpretation is deferred to Section 4, where we link estimated utility differences to revenues from TfL route procurement contracts.

if operator i were to enter; and the centrality of garage j in operator i's own garage network upon entry. The error term ϵ_{ij} varies randomly across operators and garages. It aggregates idiosyncratic route operation cost draws v_{ijr} from (1) across all routes near garage j, as well as measurement error arising from the low-dimensional representation (G) of the network variables (Γ). The deterministic part of the garage utility function is denoted $f(\cdot)$. Next, we discuss our main specification; alternative specifications are considered in Section 3.4.

The variable $YearsInc_j$ captures *persistence* in this market arising from the multi-year nature of route contracts, with θ_1 measuring the magnitude of this advantage per additional year of tenure. $Nonvacant_i$ is an intercept common to all active operators capturing the differential value of the garage plot when used for bus operations relative to alternative uses—that is, the average utility relative to *the market* as outside option. $\log(NrGarages)_i$ captures potential scale effects in the total number of garages an operator owns, with diminishing returns to garage expansion: each additional garage contributes less to operator utility as the network grows larger. $RouteStarts10Min_j$ is the number of route endpoints within a 10-minute drive from garage j, capturing *dead miles*: a higher value implies less empty driving time between the garage and nearby routes, regardless of who operates them. $MindurOwnGar_{ij}$ is the minimum drive time between garage j and any other garage owned by operator i, capturing *economies of density* net of any business-stealing effects. $CompGar10Min_{ij}$ is the number of competing operators' garages within a 10-minute drive from garage j, capturing *local monopoly rents*.

Some garages are not yet developed at the start of the sample in 1995, some remain temporarily vacant, and a few are permanently closed before 2019. To avoid modeling garage development and to reflect the highly constrained London property market, the garage set $\mathcal{J}$ is held constant over time and contains all garages ever in use during the sample period. When a garage is not owned by a bus operator it is assigned to *the market*—the outside option $i = 0$, always included in $\mathcal{N}$—representing the garage plot at its alternative use value. Since the level of $\tilde{\Pi}_{ij}$ is not identified, we normalise the mean utility of the outside option to zero and interpret coefficients accordingly.

The estimation strategy compares $\tilde{\Pi}_{ij}(\Gamma)$ across operators i for the same garage j, so that garage-specific variables cancel from the choice probabilities and their coefficients are not separately identified. We nonetheless include $RouteStarts10Min_j$ and $YearsInc_j$ in (12) by normalizing them to zero for the outside option. This normalization rests on two identifying assumptions.

First, the alternative use value of a garage plot does not depend on its proximity to bus routes: a vacant plot repurposed for logistics or light industrial use has no direct need for bus route access. The assumption requires that the market values the plot for its general locational attributes—size, road and rail access, central location—rather than bus proximity specifically. If central plots happen to be close to bus routes and the market values them partly for this reason, θ_4 would be attenuated. Second, the incumbency advantage from tenure is specific to bus operations. Bus-operation-specific investments—tailored maintenance facilities, driver logistics, de-

pot infrastructure—depreciate over time and are of limited relevance to an alternative use of the plot, so that $YearsInc_j = 0$ for $i = 0$ is appropriate. Under both assumptions, θ_4 and θ_1 are identified from a difference-in-differences argument: each coefficient measures how much more likely a garage is to remain occupied by a bus operator—rather than revert to the market—when it has low dead miles or was recently acquired, relative to what general locational attributes alone would predict. By contrast, garage capacity and transaction prices are garage-specific constants that enter identically for operators and the market alike and are therefore not identified from utility differences in the reversed-choice framework.[19]

A remaining concern is whether unobserved garage-level characteristics could bias the network variable estimates. The identifying assumption is that, conditional on the observed network variables, any remaining unobserved heterogeneity across garage locations is uncorrelated with the operator-specific regressors.

We verify empirically that the garage value function $\tilde{\Pi}(\cdot)$ satisfies supermodularity, confirming that the maintained assumptions for existence of a pure-strategy Nash equilibrium are consistent with the data ($\hat{\theta}_5 \leq 0$ and $\hat{\theta}_6 \leq 0$ as documented in Table 5). Our approach exploits the exclusivity of garage use and the temporal structure of our data by estimating on observed ownership *transitions* rather than the full cross-sectional network, varying the number of garages assumed to choose operators simultaneously as a robustness check on equilibrium selection.

2.3.1 Sequential discrete choice

We now re-introduce the time dimension and use t subscripts where variables are time-varying. In the vast majority of cases only one garage changes ownership at a given changedate (Table 1); we therefore first estimate the model under a sequential discrete choice framework with myopic decision-makers, treating each ownership change as an independent event. Cases where multiple garages change owner on the same changedate are treated as sequential draws from the same choice set, ordered as observed in the data. The observed garage ownership vector at each point in time is interpreted as a Nash equilibrium of the garage-chooses-operator game: conditional on the rest of the network remaining fixed, no garage has a profitable unilateral deviation to a different operator. This is the natural equilibrium concept for a static game, and is satisfied by construction when each garage is assigned to its highest-value operator given the observed assignments of all other garages.[20]

[19] One practical implication is that garage transaction prices need not be observed or estimated to determine the counterfactual network allocation, since they cancel from the equilibrium assignment condition as shown in Section 2.2. The same goes for garage capacity, although this is observed for two-thirds of the garages. The assumption that capacity is orthogonal to the operator-specific network variables is verified empirically in Appendix B: regressing capacity on the network variables using garage-level averages across 61 garages with available data yields an adjusted R^2 of 0.008, and neither *MindurOwnGar* nor *CompGar10Min* are significantly correlated with capacity.

[20] In other words, when garage j selects an operator at time t according to (11), it conditions on the observed state of the network at that point ($\Gamma_t \equiv \mathbf{a}_{-jt}$).

Adding the distributional assumption $\epsilon_{ijt} \overset{i.i.d.}{\sim} F_\epsilon(0, \sigma)$,[21] the choice probability for garage j selecting operator i at time t is

$$\begin{aligned} P_{ijt}(\theta) &= \Pr\Big(f(X_{ijt}, G_{ijt}; \theta) + \epsilon_{ijt} \geq f(X_{hjt}, G_{hjt}; \theta) + \epsilon_{hjt} \quad \forall\, h \in \mathcal{N}_t\Big) \\ &= \int \mathbf{1}\Big[f(X_{ijt}, G_{ijt}; \theta) + \epsilon_{ijt} \geq f(X_{hjt}, G_{hjt}; \theta) + \epsilon_{hjt} \quad \forall\, h \in \mathcal{N}_t\Big]\, dF_\epsilon \end{aligned} \tag{13}$$

where $\mathcal{N}_t$ is the set of active operators with at least one garage in the market at time t, always including the vacant choice $i = 0$. By standard arguments, for any fixed (X_t, G_t, θ), the utility-maximising choice is unique with probability one, establishing uniqueness of garage-operator matches nonparametrically under mild conditions on F_ϵ. When further specifying F_ϵ to be the standardized Type I extreme value distribution — which satisfies the conditions for uniqueness by being continuous and hence atomless —, as we do in estimation, the integral in (13) reduces to the convenient closed form (McFadden, 1984):

$$P_{ijt}(\theta) = \frac{\exp\Big(f(X_{ijt}, G_{ijt}; \theta)\Big)}{\sum_{h \in \mathcal{N}_t} \exp\Big(f(X_{hjt}, G_{hjt}; \theta)\Big)} \tag{14}$$

The logit functional form sets the scale parameter of ϵ to one, fixing the implied error variance at $\pi^2/6$, so estimated coefficients are scaled by $\sqrt{\pi^2/6}/\sigma$ relative to the true utility coefficients, where σ is the standard deviation of the true unexplained component. The finite-dimensional parameter vector θ is estimated by maximum likelihood, using the observed identity of the operator entering each garage at each changedate and the network variables at that date.

The estimation strategy requires that the event of an ownership change is independent of $f(\cdot)$ conditional on the observables. This would be violated if firms make forward-looking entry decisions, for instance entering a location even at a short-run loss to preempt future competitor expansion in, e.g., (Goolsbee and Syverson, 2008; Zheng, 2016). We recognize that this is a limitation, shared with prior entry literature (e.g., Jia, 2008; Ellickson, Houghton and Timmins, 2013; Nishida, 2015). To understand the role of the purely static timing assumption, we also estimate the model based on the utility of *bundles* of two consecutive ownership changes, as addressed next.

[21] The location normalization $\mu = 0$ is without loss of generality: since ϵ_{ijt} is drawn i.i.d. from $F_\epsilon(\mu, \sigma)$ for all $i \in \mathcal{N}_t$ including the vacant option $i = 0$, a common location μ cancels from every utility difference that determines choices and is therefore unidentified. This is distinct from the normalization of the vacant option's deterministic utility to zero, which pins down the level of $f(\cdot)$ and identifies the non-vacant intercept in X.

2.3.2 Sequential bundled choice

The bundle framework is the natural first extension toward dynamics: by treating two sequential garage transactions as a joint decision, the entry of the first operator internalizes its consequences for the subsequent transaction, mimicking a one-step-ahead lookahead. Extending this to three or more steps is computationally prohibitive—the number of bundle combinations grows from 1,841 under sequential choice to 33,513 under two-step bundling—so we treat the two-step bundle as a tractable approximation. Comparing θ and θ^B serves as a diagnostic: if the spatial rent estimates are stable across the two specifications, this supports the static framework as a valid simplification; it would indicate longer planning horizons.

Let $B = \{(i_1, j_1), (i_2, j_2)\}$ denote the bundle in which operator i_1 enters garage j_1 and, in the subsequent transaction, operator i_2 enters garage j_2. Here i_1 and i_2 may be the same operator entering different garages in sequence; j_1 and j_2 may coincide only if $i_1 \neq i_2$, since exclusivity precludes the same operator entering the same garage twice consecutively.

The value of bundle B is the average of the deterministic utility when i_1 enters j_1 and the deterministic utility when i_2 enters j_2 *conditional on* i_1 *having entered* j_1, plus a bundle-specific unobservable $\epsilon^B \sim^{i.i.d.} F_{\epsilon^B}(0, \sigma^B)$ of type-1 extreme value:

$$\begin{aligned}
\tilde{\Pi}^B(\Gamma) = \; & \theta_1^B \left(\text{YearsInc}_{j_1} + \text{YearsInc}_{j_2}\right)/2 \\
& + \theta_2^B \left(\text{Nonvacant}_{i_1} + \text{Nonvacant}_{i_2}\right)/2 \\
& + \theta_3^B \left(\log(\text{NrGarages})_{i_1} + \log(\text{NrGarages})_{i_2}(i_1)\right)/2 \\
& + \theta_4^B \left(\text{RouteStarts10Min}_{j_1} + \text{RouteStarts10Min}_{j_2}\right)/2 \\
& + \theta_5^B \left(\text{MindurOwnGar}_{i_1 j_1} + \text{MindurOwnGar}_{i_2 j_2}(i_1 j_1)\right)/2 \\
& + \theta_6^B \left(\text{CompGar10Min}_{i_1 j_1} + \text{CompGar10Min}_{i_2 j_2}(i_1 j_1)\right)/2 + \epsilon^B
\end{aligned} \tag{15}$$

Three network variables are evaluated conditional on i_1 entering j_1: $\mathit{MindurOwnGar}_{i_2 j_2}(i_1 j_1)$ is the minimum drive time between j_2 and any garage of operator i_2 given i_1's entry; $\mathit{CompGar10Min}_{i_2 j_2}(i_1 j_1)$ is the number of competing garages within 10 minutes of j_2 for operator i_2 given i_1's entry; and $\mathit{log(NrGarages)}_{i_2}(i_1)$ is the natural logarithm of the number of garages of operator i_2 other than j_2, given i_1's entry. The *Nonvacant* terms enter additively and are set to zero for the outside option $i = 0$, so the bundle intercept equals θ_2^B when both garages are assigned to active operators, $\theta_2^B/2$ when one is assigned to an active operator and the other to the market, and zero when both are vacant—the outside option bundle. As above, all remaining variables are set to zero for the outside option bundle. The bundle choice probability is

$$P_{Bt}(\theta^B) = \frac{\exp\Big(f(X_B, \Gamma_B; \theta^B)\Big)}{\sum_{c \in \mathcal{B}_t} \exp\Big(f(X_c, \Gamma_c; \theta^B)\Big)} \tag{16}$$

where $\mathcal{B}_t$ collects all possible bundles $(i_1, i_2) \in \mathcal{N}_t \times \mathcal{N}_t$ (with replacement), with j_1 fixed as the garage changing ownership at t and j_2 as the next garage to transact.

Even if firms make only single-garage decisions throughout the sample, the bundle estimates θ^B may differ from θ in levels due to differences in the implied error variance $\sigma \neq \sigma^B$, which scales the population coefficients. Since this scaling is common to all coefficients within a model, ratios of coefficients — and hence the implied trade-offs between spatial rent determinants — are scale-invariant and directly comparable across the two specifications, which we focus on in Section 3.2.

3 Estimation results

This section presents the estimation results of the *garages choose operators* model, resting on the assumption that when a garage changes ownership, the entrant is the operator with the highest utility for that garage among all active operators given the state of the network at that time.

3.1 Baseline estimates

Summary statistics of the relevant variables are presented in Table 4 and the main estimation results in Table 5. We discuss each component of the garage utility function in turn based on the baseline sequential estimates, before turning to a comparison with the bundle estimates in Section 3.2.

Incumbency benefit. The coefficient on *YearsInc*—the number of years the incumbent has been in the garage—is positive and significant ($\hat{\theta}_1 = 0.121$, s.e. 0.040, $p < 0.01$). The linear specification implies a constant rate of accumulation of incumbency advantage over time: each additional year of tenure increases the incumbent's utility for the garage at a constant rate, reflecting the gradual build-up of garage-specific investments and operational logistics that are costly for a new operator to replicate.[22]

Alternative market usage. Active operators have significantly higher mean utility for a garage than the outside option ($\hat{\theta}_2 = 1.886$, s.e. 0.529, $p < 0.001$), confirming that bus operators value garages substantially more than their alternative use value, consistent with the garage being a productive asset specifically tied to TfL route procurement.

[22] An alternative parameterization using $\exp(-YearsInc_j)$ is arguably more naturally motivated by the five-year contract structure: if an operator enters a garage to serve one or more contracts immediately after winning, the incumbency advantage would be highest at entry and decay toward zero over the contract duration. However, the robustness checks in Section 3.4 show that the linear specification fits the data better. This may reflect the staggered nature of route contracts—whereby an incumbent simultaneously holds contracts at different stages of their duration—so that the garage retains residual value throughout the tenure rather than only at the start.

Scale effects. The coefficient on $\log(NrGarages)$—the log of the total number of garages operated by the firm—is negative and highly significant ($\hat{\theta}_3 = -1.135$, s.e. 0.124, $p < 0.001$). The logarithmic specification implies diminishing returns to garage expansion: each additional garage contributes less to operator utility as the network grows larger. Conditional on network centrality and competitor proximity, larger operators derive less marginal utility from an additional garage, consistent with firms having fewer uncovered routes to serve from a new garage as their network becomes denser.[23]

Dead miles. The coefficient on *RouteStarts10Min*—the number of route endpoints within a 10-minute drive—is positive as expected but statistically insignificant in the baseline specification ($\hat{\theta}_4 = 0.030$, s.e. 0.028). This is consistent with the reduced-form evidence in Section 1.2, where the dead miles coefficient was similarly attenuated once strategic network controls were included. In the baseline, the dead miles variable may be partially collinear with the non-vacancy intercept, both of which partly reflect garage centrality in the route network. The bundle model, by internalizing the network consequences of sequential entry, provides better identification of this component: the bundle estimate is large and highly significant ($\hat{\theta}_4^B = 0.161$, s.e. 0.023, $p < 0.001$). We therefore treat the baseline estimate as a conservative lower bound on the true dead miles effect while noting that both specifications agree on the sign and that the dead miles coefficient does not affect the robustness of the other spatial rent estimates, which are stable across both specifications.

Economies of density. The coefficient on *MindurOwnGar*—the minimum drive time to the operator's nearest own garage—is negative and highly significant ($\hat{\theta}_5 = -0.091$, s.e. 0.010, $p < 0.001$). Garages are more valuable when they are closer to the operator's existing network, confirming the presence of economies of density in garages. These provide a cost-based rationale for the observed spatial clustering of operators in certain parts of London.

Local monopoly rents. The coefficient on *CompGar10Min*—the number of competitor garages within a 10-minute drive—is negative and significant ($\hat{\theta}_6 = -0.334$, s.e. 0.110, $p < 0.05$). Garages are less valuable when surrounded by competitor garages, consistent with the local monopoly rent mechanism: proximity of rivals compresses equilibrium mark-ups in route auctions, reducing the expected profit from operating routes nearby. To quantify the relative importance of economies of density and local monopoly rents, we compare the ratio $|\hat{\theta}_6|/|\hat{\theta}_5| = 3.7$, which gives the reduction in minimum own-garage drive time (in minutes) that offsets one additional competitor garage within 10 minutes.

[23] The logarithmic specification fits the data better than a linear specification in levels, as shown in the robustness checks of Section 3.4.

Table 5: Structural estimates of garage-operator utility parameters

	baseline	bundle
Incumbency benefit (YearsInc)	0.121**	0.189***
	(0.040)	(0.033)
Garage is not vacant (Nonvacant)	1.886***	0.813**
	(0.529)	(0.277)
Log nr. own garages (log(NrGarages))	−1.135***	−1.574***
	(0.124)	(0.121)
Nr. bus route starts within 10 min. drive time (RouteStarts10Min)	0.030	0.161***
	(0.028)	(0.023)
Min. drive time to any own garage (MindurOwnGar)	−0.091***	−0.120***
	(0.010)	(0.008)
Nr. comp. garages within 10 min. drive time (CompGar10Min)	−0.334**	−0.427***
	(0.110)	(0.109)
Number Observations	1,841	33,513
Number of Choice Situations	192	381
Own garage proximity (mins) offsets extra comp. garage	3.668	3.552
Pr(highest-utility operator correctly predicted)	0.505	0.132
Mean AUC	0.868	0.875
McFadden's pseudo-R^2 w.r.t. Null Model (ρ_0^2)	0.554	0.422

Notes: Statistical significance: + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001. The column "baseline" contains the structural estimates from the sequential single-move discrete choice model based on equation (12). The column "bundle" contains the model estimates when bundles of two garages and operators are chosen sequentially, based on the utility function in equation (15). Summary statistics are in Table 4.

Table 6: Testing structural estimate equality across baseline and bundle specifications

	baseline	bundle	p-value
Ratio $\hat{\theta}_k/\hat{\theta}_6$ where $\hat{\theta}_6$ = CompGar10Min			
Incumbency benefit (YearsInc)	-0.362	-0.443	0.688
Garage is not vacant (Nonvacant)	-5.647	-1.907	0.156
Log nr. own garages (log(NrGarages))	3.399	3.689	0.846
Nr. bus route starts within 10 min. drive time (RouteStarts10Min)	-0.091	-0.377	0.019
Min. drive time to any own garage (MindurOwnGar)	0.273	0.282	0.945

Notes: Ratios between the coefficients on the listed variable to the coefficient on *CompGar10Min* in the relevant specifications reported in Table 5. Ratios are scale-invariant and directly comparable across specifications despite differences in implicit error variance σ. The p-value is from a two-sided z-test of equality of the two ratios, as defined in (17), with standard errors obtained via the delta method (Greene, 2012) applied to the respective maximum likelihood covariance matrices.

3.2 Bundle choice estimates

As discussed in Section 2.3.2, the bundle model allows operators to internalize the network consequences of sequential entry, providing a robustness check on the static baseline. Since the two

specifications differ in their implicit error variance σ, coefficient levels are not directly comparable across columns. We therefore assess robustness by comparing coefficient ratios relative to $\hat{\theta}_6$ (the competitor garage coefficient), as such ratios are scale-invariant and directly interpretable as trade-offs between garage value determinants.

To formally test whether these ratios differ across specifications, we apply the delta method to obtain standard errors (Greene, 2012) for each ratio within each model and construct a z-test of equality exploiting the independence of the two estimators:

$$z_k = \frac{\hat{\theta}_k/\hat{\theta}_6 - \hat{\theta}_k^B/\hat{\theta}_6^B}{\sqrt{\widehat{\text{Var}}(\hat{\theta}_k/\hat{\theta}_6) + \widehat{\text{Var}}(\hat{\theta}_k^B/\hat{\theta}_6^B)}} \tag{17}$$

for $k = 1:5$. Variances are obtained via the delta method applied to the respective maximum likelihood covariance matrices. The results are reported in Table 6.

The test fails to reject equality for four of the five coefficients, including the key trade-off between economies of density and local monopoly rents ($\hat{\theta}_5/\hat{\theta}_6$: baseline 0.273, bundle 0.282, $p = 0.945$). This is the primary object of interest—the number of minutes of own-garage proximity that offsets one additional competitor garage—and its near-identical value across specifications provides strong support for the baseline estimates. Scale effects and the incumbency benefit are likewise statistically indistinguishable across models ($p = 0.846$ and $p = 0.688$ respectively). The one ratio that does differ significantly across specifications is that of dead miles ($p = 0.019$): the bundle estimate of *RouteStarts10Min* is large and significant (Table 5) while the baseline is not, so the ratio to *CompGar10Min* is roughly four times larger in the bundle model. As discussed above, this reflects better identification of the dead miles component in the bundle model rather than instability in the spatial rent estimates (local monopoly rents and economies of density are stable across specifications).[24]

Taken together, these results support the static sequential framework as a valid simplification for the purpose of identifying the spatial rents in this environment. Extending the planning horizon from one to two steps does not materially alter the picture of how spatial positioning drives garage value in this market. At the same time, we cannot rule out that alternative bundling structures—for instance, non-sequential bundles or three-step lookaheads–would yield different estimates, particularly for the dead miles component where the two specifications do differ. What the comparison does establish is that a parsimonious static model without prices or explicit dynamics is able to recover stable and economically interpretable estimates of the two central forces—economies of density and local monopoly rents—that theory predicts should govern location choice in markets with spatial competition.

[24]Capturing dead miles better with the two-garage model partially motivates why we include more detailed operator-garage specific dead mile measures based on simulated route assignments in our counterfactuals, based on equation (5), noting that route-garage assignment is not observed throughout history.

3.3 Model fit

Multinomial choice models do not admit a single summary fit statistic analogous to R^2. We therefore assess fit using the mean Area Under the ROC Curve (AUC), averaged across operators using a one-vs-all classification scheme and weighted by operator frequency in the sample.[25] The baseline model achieves a mean AUC of 0.868 and a McFadden pseudo-R^2 of 0.554; the bundle model achieves an AUC of 0.875 with a pseudo-R^2 of 0.422, the lower value reflecting the much larger choice set rather than worse fit.[26] The AUC metric, which is robust to the number of alternatives, confirms that both specifications have strong discriminatory power and that the key spatial rent parameters are stable across estimation approaches. The model correctly identifies the highest-utility operator in 50.5 percent of all choice situations, out of an average of 10.75 alternatives per situation—a substantial improvement over random assignment.

Figure A5 compares in-sample predicted and observed entry frequencies, computed as the model-implied average probability that each operator is selected across all choice situations against the observed share of ownership changes in which that operator entered a garage. The model tracks individual operator entry patterns closely across the distribution. The main discrepancy is an over-prediction of vacancy: the model assigns garages to the outside option in 38.5 percent of cases against 26.0 percent in the data. This reflects the difficulty of capturing the timing of garage closures and developments, which often occur for reasons exogenous to bus operations, and supports the modeling choice to treat the set of garages as the fixed outer set of ever-occupied plots rather than modeling entry and exit of garage infrastructure explicitly.

A second discrepancy highlighted in Figure A5 is that the model under-predicts entry probabilities for Arriva and Go-Ahead. This follows mechanically from the estimated diseconomies of scale in $\log(NrGarages)$ combined with these operators' large garage holdings (Figure A3): conditional on network centrality and competitor proximity, the model assigns low marginal value to further expansion by already large operators. Appendix C explores a complementary explanation, motivated by the UK Competition Commission's concerns about geographic market segregation in local bus markets: whether unrealized garage transactions that the model predicts to be profitable are consistent with collusive market-sharing. These unrealized transactions concentrate on garages spatially isolated from competitor networks—the local monopoly rent variables of Table 5—but their incidence rises rather than falls with the number of active firms, contrary to the market-sharing prediction. Notably, reallocating garages to their highest-value operators shrinks Arriva to a fringe player while expanding GoAhead, ComfortDelGro,

[25] The AUC measures the probability that the model assigns a higher predicted probability to the true entrant than to a randomly drawn non-entrant, ranging from 0.5 (random assignment) to 1 (perfect discrimination). In the multinomial setting we compute one ROC curve per operator under a one-vs-all scheme and report the frequency-weighted average following Provost and Domingos (2003).

[26] McFadden pseudo-R^2 values of 0.2–0.4 are conventionally considered to indicate excellent fit, substantially exceeding the threshold for OLS regressions of the same magnitude (Domencich and McFadden, 1975). The baseline value of 0.554 is therefore exceptional for a model with only five network variables.

and Abellio, mirroring the under-prediction pattern in Figure A5; whether this reflects market-sharing or firm-specific advantages outside the five network variables cannot be separated, and we claim no evidence of collusion.[27] Here, we simply conclude that the spatial rent model fits the observed network changes remarkably well given its parsimony, without explaining specific operator-specific deviations from its predictions.

3.4 Alternative specifications

We assess the robustness of the baseline estimates to alternative specifications. First, we consider alternative specifications of the three distance components (Appendix Table A3). Replacing *RouteStarts10Min* with stopover-based dead mile measures leaves all spatial rent estimates unchanged and the stopover coefficients themselves are insignificant, confirming that the dead miles attenuation in the baseline is not specific to the start-stop measure. Replacing *MindurOwnGar* with the average drive time to own garages yields a stable local monopoly rent coefficient but a much larger trade-off ratio (9.9 vs. 3.7 in the baseline) and lower fit (AUC 0.844 vs. 0.868), reflecting the different scaling of the average relative to the minimum and confirming the minimum as the preferred summary of economies of density. Replacing *CompGar10Min* with continuous drive-time measures yields insignificant competitor distance coefficients, suggesting that the count of nearby rivals is more informative than their average distance. The baseline specification compares favorably on all fit metrics and the core spatial rent estimates are robust to the choice of distance measure.

Second, we consider alternative specifications of the incumbency and scale variables (Appendix Table A4). Replacing the linear incumbency term with the exponential form $\exp(-\textit{YearsInc})$ yields an insignificant coefficient and lower fit, confirming the preference for the linear specification. Omitting incumbency entirely leaves the spatial rent estimates and fit virtually unchanged, confirming that the incumbency specification does not drive the primary findings. Replacing $\log(\textit{NrGarages})$ with *NrGarages* in levels reduces fit noticeably while leaving the trade-off ratio stable; omitting the scale variable entirely reduces fit further without materially affecting the spatial rent estimates. Omitting the *Nonvacant* intercept restores significance of the dead miles coefficient, further supporting the collinearity argument in Section 3. Throughout, the trade-off ratio between economies of density and local monopoly rents ranges from 3.1 to 4.7, and the core finding that these two forces are the primary determinants of garage value is robust across all alternative specifications considered.

[27] The spatial rent estimates are identified entirely from observed ownership transitions, without procurement bid data, and are agnostic about competitive conduct at the auction stage. The only assumption required is that transactions that do take place allocate garages to the operator that values them most, making the appendix exercise out-of-sample relative to the estimation.

4 Counterfactuals: implications for urban planning

The structural estimates link garage value directly to spatial positioning in the network of routes and competing garages. We exploit this to assess two urban planning questions: where in London would new garage infrastructure generate the most private value to operators, and where are capacity constraints on existing garages most binding? The first question is addressed descriptively using the estimated spatial rent surface across candidate locations; the second requires the bus-garage assignment equation, which we validate against observed tender data before applying it to the capacity expansion counterfactuals.

4.1 Network quality of candidate garage locations

For each point ℓ on a 500-meter resolution grid covering Greater London, we compute the three spatial rent components of the reduced-form garage utility function (12) as if a new garage were located at ℓ.[28] For each candidate location ℓ and each operator i, we then evaluate $\hat{\Pi}_{i\ell}(\Gamma_{-\ell})$ using the baseline structural estimates, excluding any garages situated inside the 500 m^2 area ℓ but holding the rest of the 2019 garage-operator network Γ fixed. We also determine the highest-value operator at each location, as [29]

$$i^*(\ell) = \arg\max_i \hat{\Pi}_{i\ell}(\Gamma_{-\ell})$$

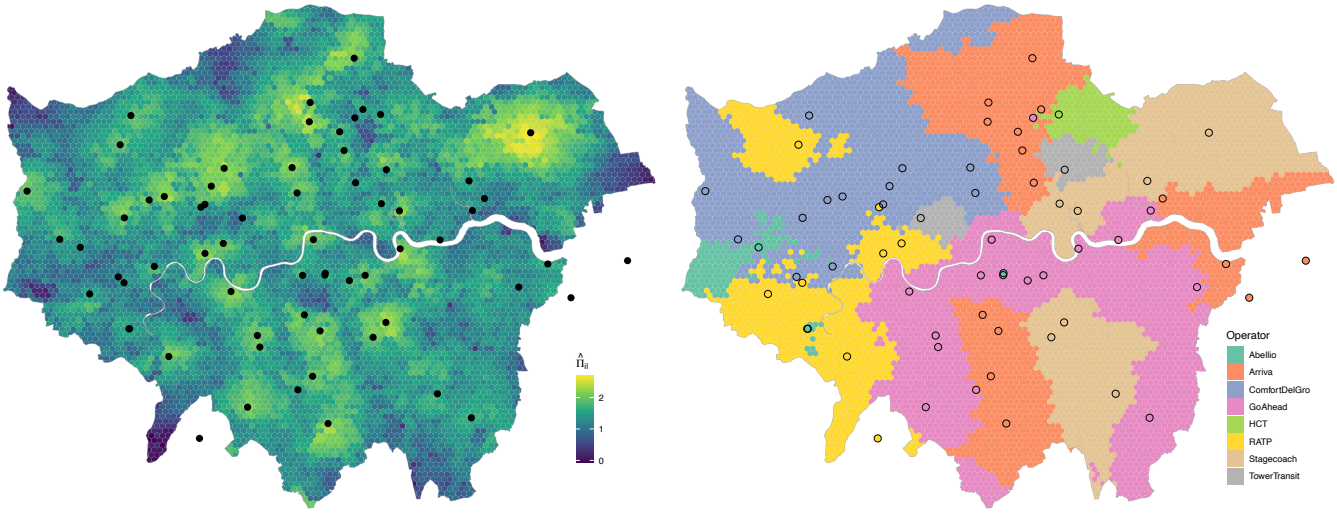
This exercise is purely descriptive: it identifies where in London the marginal value of additional operating capacity is largest for each operator, based on the spatial positioning of candidate locations relative to routes and existing garages. It does not use the bus-garage assignment in equation (5) and abstracts from garage capacity—questions we take up in Sections 4.2–4.3.[30] Figure 2 presents the two complementary views of the spatial rent surface. Panel (a) shows the overall *private* value of releasing land for garage development at each candidate location, independently of which operator would benefit. Rather than a clear geographic gradient, the surface reveals many dispersed high-value pockets scattered across London, reflecting that the 2019 garage network is broadly well-positioned relative to the route network—each operator has already claimed the most valuable nearby locations, leaving isolated pockets of residual private value where the network remains sparse. The northeast is the main exception, where a larger contiguous area of higher private value suggests that garage coverage is genuinely underprovided relative to the density of the route network there.

[28] The relevant variables are the number of route endpoints within a 10-minute drive, the minimum drive time from ℓ to the nearest existing garage of each operator i, and the number of competitor garages within a 10-minute drive of ℓ. Drive times are computed using the OSRM routing engine on the London road network.

[29] In what follows, $\hat{\Pi}$ denotes the estimated deterministic garage utility given our baseline estimates.

[30] Incorporating bus-garage assignment can address issues like the possibility that a garage in an otherwise unserved area could generate stand-alone value by accessing routes not currently viable from existing garages.

Figure 2: Network quality surface for new garage locations



(a) Location value surface

(b) Optimal entering operator

Notes: Panel (a) shows the highest predicted utility across all operators on a 500-meter grid, reflecting the overall value of releasing land for garage development at that location. Panel (b) shows the operator that would benefit most from a new garage at each location. Dots mark existing 2019 garages. Both panels are evaluated using the 2019 garage-operator network Γ and reflect pure network position value, abstracting from garage capacity and bus-garage assignment.

Panel (b) maps $i^*(\ell)$, the highest-value operator at each location. The spatial segmentation is clean and contiguous: Stagecoach dominates large areas in the south and south-east, Abellio the north-west, and smaller operators occupy distinct central territories. Most existing garages lie within their own operator's preferred region, confirming that the current network broadly reflects spatial rent optimization. At a high level, the results reveals where operators would have the strongest incentive to lobby for the release of industrial land or planning permissions for new garage development, with the northeast emerging as the area where private returns to new garage infrastructure would be largest.[31]

4.2 Private returns to garage capacity expansion

This section quantifies the returns to expanding capacity at existing garages with 50 buses—approximately half of the mean capacity in the data—and asks how much private profit the owning operator would gain. The answer combines two components capturing distinct channels through which additional capacity creates value.

The first is the *structural term*. The structural model assigns each garage a deterministic valuation $\hat{\Pi}_{ij}$ reflecting its position in the operator's network: proximity to own and competitor

[31] The underlying geography is relevant: this area is situated in the middle of large stretches of designated open space, protected against development.

garages, density of nearby route starts, and incumbency tenure. Under constant returns to scale in capacity (motivated in Appendix B), dividing by the current bus count B_{jt} yields a per-bus valuation $\hat{\pi}_{ij}^{bus} = \hat{\Pi}_{ij}/B_{jt}$. This converts to monetary units via $\kappa = £4.30\text{m}$ per utility unit (24), calibrated by matching the total market value (i.e. the logit inclusive value) to observed London bus garage transaction prices, which average £ 11.2M per garage on average (see Appendix E).

The second is the *dead-miles correction*. The structural model is estimated at the garage level and does not observe route-to-garage assignments; $RouteStarts10Min$ captures route proximity only through a discrete route count. When capacity expands, the assignment technology of equation (5) predicts which additional routes are drawn in by proximity. These marginal routes need not share the dead-mile profile of existing assignments: if closer, the operator saves on empty driving; if further, costs rise. The correction compares the dead-mile cost of each marginal route m' to the PVR-weighted baseline average $\overline{DM}_j$, scaled by the estimated dead-mile penalty $\hat{\beta} = -£19{,}000$ per additional dead mile.

Combined, the private return to operator i from adding 50 buses to garage j is:

$$\Delta\hat{\Pi}_{ij}(+50 \text{ buses}) = \underbrace{\sum_{m' \in \mathcal{M}_j} \Delta b_j(m') \cdot \hat{\pi}_{ij}^{bus} \cdot \kappa}_{\text{structural term}} + \underbrace{\hat{\beta} \cdot \sum_{m' \in \mathcal{M}_j} \Delta b_j(m') \cdot \left(DM_j(m') - \overline{DM}_j\right)}_{\text{dead-miles correction}} \tag{18}$$

where $\mathcal{M}_j$ is the set of marginal routes drawn in beyond the current assignment, ordered by proximity; $\Delta b_j(m')$ is the number of additional buses from route m' stationed at the garage (equal to the full PVR of m' for all but the last marginal route, truncated so the total increment is exactly 50 buses); and all other terms are as defined above. Further details are in Appendix E.

Figure 3 presents the joint distribution of the two components across garages. The horizontal axis measures structural per-bus value in monetary units, recentered relative to the lowest-valued garage.[32] The vertical axis measures the dead-miles correction, negative when marginal routes are further than the baseline average and positive when closer.[33] For most garages the correction is negative—marginal routes are further than the baseline average—underscoring the increasing marginal cost of expansion as closer routes are already served. This implies that even absent land constraints, dead-mile costs alone favor multiple smaller garages over a single large depot. Garages like Bow and Tottenham face severe negative corrections: marginal expansion requires serving distant routes, multiplying dead-mile costs and eroding expansion value. Garages such as Waterloo and Harrow are similar in structural value but retain nearby unserved routes, avoiding this penalty. Erith and Kangley Bridge Road combine high structural

[32] The lowest-valued garage is Grays, the easternmost garage in the sample, located just beyond the Greater London boundary and owned by Arriva.

[33] The correlation between the two terms is positive ($r = 0.161$): structural estimates capture that garages closer to more routes have higher value regardless of current assignments, while variation in the dead-miles penalty reflects route-specific empty driving times beyond the discrete *RouteStarts10Min* route count.

value with proximity to unserved routes, making them optimal candidates for investment.

To explain what drives private returns, we regress $\Delta\hat{\Pi}_{ij}$ on structural covariates, capacity, utilisation, and baseline dead miles, with and without operator fixed effects (Table A5). Average operator-garage targeted dead miles (before expansion) is the dominant cost driver: garages serving more distant routes face even higher marginal dead-mile costs on expansion.[34] Incumbency tenure is a robust positive driver, reflecting that firms stay longer in more profitable garages, and operator fixed effects absorb substantial variation—R^2 rises from 0.782 to 0.915.

The key policy-relevant finding concerns competitor proximity. One additional competitor garage within a 10-minute drive reduces the return to a 50-bus expansion by £0.9M—approximately 10 percent of the baseline value, increasing to £1M conditional on operator fixed effects. Deflating $\Delta\hat{\Pi}$ by borough-level median house prices even slightly stronger results (Columns (4)–(6) of Table A5). This minimizes the concern that the competitor proximity effect is driven by competitor-sparse areas simply being cheaper. When deflating by ward-level median house prices; the competitor-proximity effect strengthens to −£1.4M–1.6M (Columns (7)–(9)), suggesting that, if anything, the borough-level adjustment understates the spatial-isolation incentive by missing within-borough land-value gradients. Taken together, the stability and strengthening of this coefficient across specifications minimises the concern that the result reflects unobserved fixed expansion costs at peripheral locations.

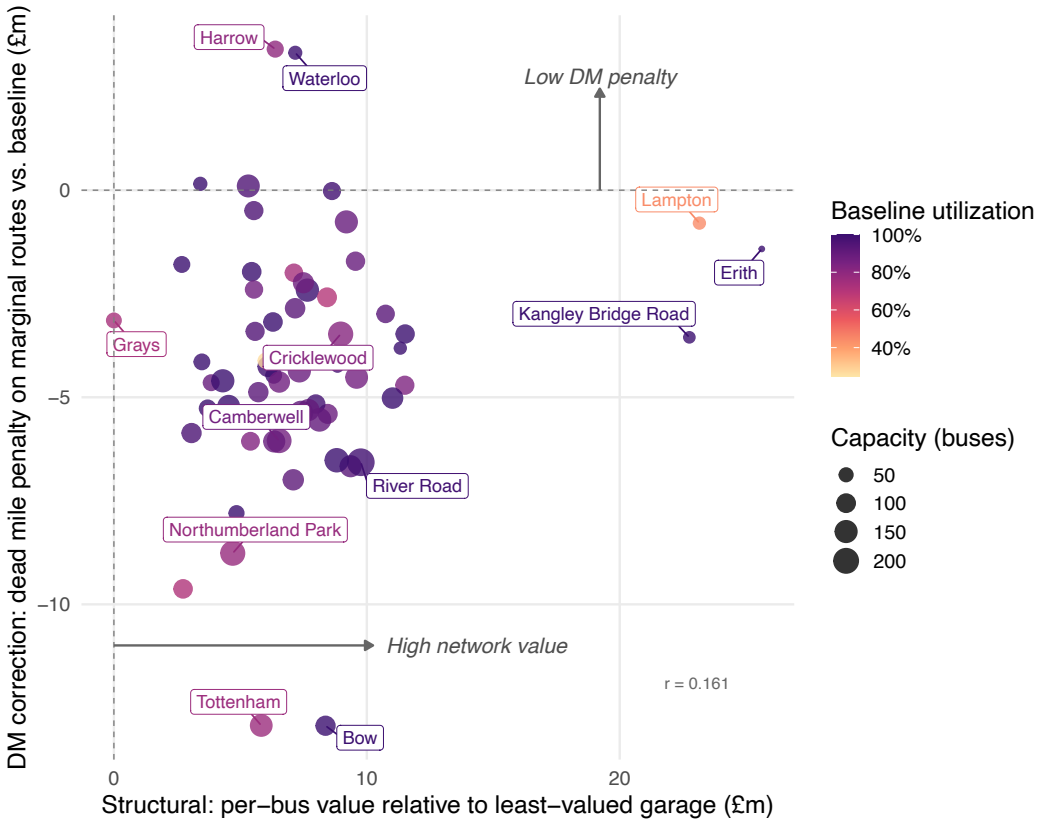
The results of these capacity expansion counterfactuals quantify the wedge between private and social returns: operators are incentivised to protect spatial isolation, generating local monopoly rents at the expense of procurement costs borne by the taxpayer. Regulation has value precisely because this wedge is not internalised by operators.

4.3 Welfare-maximizing global allocation

A social planner seeking to minimise local monopoly rents faces the problem of reassigning existing garages across operators, holding the route and garage network fixed. The primary inefficiency arises because operators have a private incentive to maintain spatial isolation from competitors: by locating away from rivals, they sacrifice proximity to routes and own garages, increasing dead miles and the associated externalities borne by society. A social planner therefore seeks to minimise these rents, while holding operational costs no higher than at the 2019 baseline —thereby not artificially lowering operator profits by increasing costs. Formally, the

[34] Targeted dead miles are not included in the structural model as they are only observed for garages in the final allocation. The two measures are collinear, which explains the negative coefficient on *RouteStarts10Min* in columns (1)–(3), and the attenuation of that coefficient when baseline dead miles are excluded (column 3), though it remains statistically insignificant across specifications.

Figure 3: Private returns to capacity expansion



Notes: Structural term ($\hat{\pi}_{ij}^{bus}$, horizontal axis) and dead miles correction (vertical axis) from returns to capacity expansion (18), for all 62 garages in the sample with observed capacity that are not vacant and have buses stationed in 2019. Garages are colored by baseline capacity utilization and sized by capacity. Labeled garages rank among the three highest on the structural term, the two most extreme on the dead miles correction, or the four largest by capacity. Grays serves as a reference point with lowest $\hat{\pi}_{ij}^{bus}$.

planner solves[35]

$$W_1 = \max_{\Gamma'} \; -\sum_j \left[\hat{\Pi}_{i_j,j}(\Gamma')\right], \tag{19}$$

$$\text{subject to:} \quad \sum_j \text{MinDurOwnGar}_{i_j,j}(\Gamma') \leq \sum_j \text{MinDurOwnGar}_{i_j,j}(\Gamma), \tag{20}$$

$$\sum_{i \in \mathcal{I}} \mathbf{1}(\text{operator } i \text{ exits under } \Gamma') = 0, \tag{21}$$

$$\sum_j \log(\text{NrGarages})_{i_j,j}(\Gamma') \leq \sum_j \log(\text{NrGarages})_{i_j,j}(\Gamma), \tag{22}$$

[35]Garage exclusivity is maintained, so some local monopoly rents remain. We explore a more drastic *unbundling* policy in Marra and Oswald (2024), but recognize that full fragmentation of operators across garages involves additional unmodeled frictions (placement of logistics and canteens, parking within the garage, entry-exit congestion).

Table 7: Welfare effects from garage reallocation

	I: No reorganization costs				II: Linear reorganization costs			
	W_1	W_2	W_3	W_4	W_1	W_2	W_3	W_4
Number of garage reallocations	64	25	22	33	12	2	3	9
Welfare gain (% change)	22.1%	15.5%	11.3%	17.4%	6.0%	5.5%	1.5%	4.2%
Main welfare drivers:								
Local monopoly rents	24.3%	18.2%	18.1%	19.6%	16.2%	7.1%	5.7%	9.1%
Dead-mile externality (γ)	—	2.0%	—	—	—	−0.0%	—	—
Constrained or penalised channels (non-increasing by design):								
Increased economies of density (*)	−0.2%	−0.2%	−0.1%	−0.1%	−0.5%	−0.1%	0.0%	−0.7%
Reduced garage concentration (*)	−2.0%	−4.4%	−6.5%	−1.9%	−5.9%	−0.7%	−3.4%	−1.3%
Reduced dead mile costs (*)	—	—	−0.2%	−0.2%	—	—	0.0%	−0.0%
Sunk investment write-off	—	—	—	—	−3.7%	−0.7%	−0.8%	−2.9%

Notes: Welfare effects of planner's solution (W1), with a correction for dead mile externalities (W2), targeted garage-operator dead mile costs (W3 and W4) and an operator-specific bus fleet constraint (W4). The "linear reorganization costs" specifications penalize ownership changes (−tenure of the incumbent $\times 0.121(\hat{\theta}_1)$), listed as sunk investment write-off. The (*) entries all reflect reductions in costs across operators but enter negatively in the constrained objective functions. See Section 4.3 for further details.

where $\Gamma' \equiv \{(i'_j, j)\}_{j \in \mathcal{J}}$ denotes a feasible garage–operator assignment and $i'_j \in \mathcal{I}$ is the operator assigned to garage j under the counterfactual. Constraint (20) rules out that profits decrease through the channel of reducing own-garage economies of density. Constraint (21) rules out operators exiting the London bus market altogether and (22) rules out further concentration in garage holdings, which would both potentially increase route procurement mark-ups.[36] Because the objective depends on the full allocation through strategic network interactions the problem is a variant of the quadratic assignment problem and NP-hard in general. The feasible space of $8^{68} \approx 10^{61}$ candidate allocations rules out exhaustive search; we therefore solve it by simulated annealing with multi-restart. We take the 2019 network as benchmark, excluding vacant garages, and considers consecutive single or double garage reassignments, accepting welfare improving network changes with certainty and welfare-reducing changes with a probability that vanishes to zero as more of the state space is searched; more details are given in Appendix F.

Table 7 (column W_1) reports the results for the planner's problem defined in (19)–(22). Almost all garages (64 out of the 68 non-vacant garages) are reallocated to a different operator, achieving a welfare gain of 22.1% relative to the observed 2019 assignment. Placing more competitors in nearby garages is the main economic driver of the welfare gain. Operator costs also decrease under the optimal allocation, which generates higher economies of density and lower diseconomies of scale in number of garages, but these have a much smaller and opposite impact

[36] Constraint (22) merits further discussion. Let $n = \text{NrGarages}$. Since $\sum_j \log n_{i_j,j}$ is equivalent to $\sum_i n_i \log n_i$, and $f(n) = n \log n$ is strictly convex, this sum is minimised by equal market shares and increases sharply under concentration. The constraint therefore permits equalising transfers but blocks concentrating ones. To illustrate: with an initial distribution of $(2, 4, 8)$ garages, $\sum_i n_i \log n_i \approx 23.6$. Equalising to $(4, 5, 5)$ lowers this to 22.4 (feasible), while concentrating to $(1, 1, 12)$ raises it to 29.8 (infeasible under the constraint).

on the objective function by design. This is so because smaller costs imply greater profits, giving the algorithm a priori a reason to *increase* costs. To avoid such spurious sources of welfare gains, such proposals are constrained not to be acceptable via (20) and (22). The cost advantage of the planner's solution is therefore always non-negative but small, supposing that firms are efficient cost minimizers conditional on conflicting motives for spatial isolation.

Beyond minimising monopoly rents, a social planner should account for the external costs of empty bus movements. Internalising dead-mile externalities via a Pigouvian correction (column W_2 in Table 7) adds 2.0% of welfare gain by encouraging further dead-mile efficiency across operators' networks. These effects should be compared against a version with the same measure of dead miles implied by bus movements after garage-operator assignment, but no Pigouvian correction. Gains from reducing spatial isolation under W_2 and W_3 are very similar (18.2% versus 18.1%), but concentration falls by less under Pigouvian pricing (-4.4% versus -6.5%), as the planner reallocates garages toward large operators best placed to economize on dead miles across their network, rather than only dispersing rivals. Increasing spatial competition remains the dominant source of gains in all cases, contributing between 18.1% and 24.3% depending on the constraints imposed (W_1–W_4, Panel I), and welfare gains remain sizeable. Imposing observed 2019 bus fleet sizes as hard capacity constraints (W_4) delivers gains of 17.4%. In W_4, operators are limited to absorbing routes their existing fleet can cover, so the final network is closer to the existing one in terms of operator size.

Welfare gains from changing the garage-operator network should be evaluated against the write-off of any incumbent-specific investments. To address this, Panel II adds a social cost of incumbent displacement corresponding to the estimated incumbency benefit in the structural model. Since this penalty accumulates linearly over time, the specification is strongly status-quo favoring, permitting reallocations conservatively. This compresses gains to between 1.5% and 6.0% across all cases. Importantly, welfare gains remain strictly positive and economically meaningful, confirming that the 2019 garage allocation leaves unexploited efficiency gains even when incumbent displacement is penalized conservatively.[37]

5 Conclusions

This paper studies firm location choices and the associated spatial profit drivers (*spatial rents*) in the London bus market, using a new database of bus garage location and ownership from 1994, when the market was privatized along geographic lines. The London bus procurement setting is well-suited to this analysis: with demand tightly regulated by TfL, a garage's position

[37] Maps of the baseline and optimal allocations under each specification from Panel II are presented in Appendix Figure A6. Since the structural model contains no operator fixed effects, the identities assigned to garages in the Panel I solutions are illustrative only. Any permutation of the operator–garage assignment that preserves the vector of relative drive times and network sizes achieves the same maximum welfare; formally, the objective is invariant to relabelling of operator identities in any allocation Γ' that maintains identical network topology.

in the network of routes and competing garages is the key determinant of firm profitability. We capture three sources of spatial rents: dead-mile costs from proximity to the route network, local monopoly rents from isolation relative to competitors, and economies of density from operating routes out of nearby own garages.

The key methodological contribution is a reversed discrete choice framework that recasts the firm location problem as one where garages choose operators, yielding a tractable multinomial estimator that avoids observing transaction prices and, by conditioning on the network realized at each transition, sidesteps the dimensionality and multiplicity problems of standard entry games. The garage value function is grounded in a structural auction model linking spatial positioning to expected profitability, while remaining agnostic about competitive conduct.

The structural estimates pin down what determines the value of different locations to different firms and inform three sets of urban planning counterfactuals. The first reveals that private returns to new garage infrastructure would be largest in the northeast of London, suggesting that the private returns to releasing industrial land are largest in that area. Second, capacity expansion simulations quantify the wedge between private and social returns. The bundled garage-and-route procurement design rewards spatial isolation: each additional competitor garage within a 10-minute drive reduces the private return to a 50-bus expansion by at least £0.9M (roughly 10% of baseline value). Third, full network optimization reveals that this distortion is consequential: a planner reassigning the full garage network could increase welfare by up to 24.3% by placing competing operators in near-by garages. Local monopoly rents remain strictly positive under garage exclusivity and building constraints, and when pricing dead-mile externalities the optimal garage assignment is such that concentration falls by less, as large operators are best placed to economize on dead miles across their own network. Short-run fleet constraints roughly halve the realised gains, and a linear incumbent investment write-off introduces a status-quo bias that compresses them further—though even this most conservative specification leaves room for welfare-improving changes.

More generally, the paper shows how firm location choices in regulated urban markets encode spatial competitive structure invisible in price or quantity data. The framework applies wherever firms maintain proprietary infrastructure, capacity is geographically fixed, and proximity generates both cost synergies and competitive spillovers—conditions common to logistics, retail, utilities, and public transport. Regulators in such markets should attend not only to pricing and procurement conduct but also to the spatial organization of firm infrastructure, which can generate and entrench market power in ways that standard tools that condition on firm location are ill-equipped to address.

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